

****All exercises & prehab series mentioned here can be found on the SSDL training tab**

****If you experience pain with any of the movements, we recommend you meet with a sports medicine professional**

Test explanations & results:

- **1. Shoulder Mobility** - This assessment tests the mobility of your entire shoulder complex. Specifically the relationship of your shoulder blades sitting on the ribcage and how well your shoulder blades can move to “deliver” the ball and socket component of the shoulder. This test is graded with the following criteria:
 - Good - Your hands were within 1 fist length
 - Average - Your hands were within 2 fist lengths
 - Below Average - Your hands were outside of 2 fist lengths
- **What to do based on your results**
 - **If you scored a Good:**
 - We'd recommend being consistent with warming up the muscles of your shoulders before training to make sure you can maintain that range of motion.
 - Specifically, the Streamline Overhead Mobility and Banded Shoulder Resiliency series are our 2 go to's for warming up the shoulders before training.
 - **If you scored an Average or Below average:**
 - There is some room for improvement with your shoulders! On top of the shoulder exercises you do as part of the dryland already, it'd be a very good idea to spend something like 5-10 minutes each day doing some prehab series or exercises you feel work really well for you. Think of how many yards you swim & how many strokes you take per day at practice, all those yards & strokes need to be accounted for from a mobility standpoint.
- **Prehab Series**
 - Banded shoulder resiliency
 - Foam roller shoulder mobility
 - Shoulder mobility & resiliency
 - Streamline overhead mobility
- **Exercises**
 - T-spine mobility - Quadruped reach & rotate, t-spine open books, t-spine windmills, seated t-spine rotations, thoracic extensions w/ foam roller
 - Scapular movement - 45° wall slide, 45° wall slide to liftoff, serratus crawls, wall slides, glute bridge w/ reach, lying serratus punch, halo variations
 - Lengthening/stretching the lats - dead pullup hangs, bench lat inhibition, foam rolling/stretching the lats

- **2. Overhead Squat** - This assessment looks at how the entire body moves as one unit. Think of the body as one big chain, with each joint (shoulders, hips, ankles) being its own respective link in the chain.
 - **What to keep an eye out for:**
 - Did the athlete maintain an overhead position through the entire motion, or did their arms come forward as they descended?
 - Did the athlete's mid back round as they descended?
 - Did the athlete's knees stay in line with their toes, or come closer together?
 - Did the athlete's ankles stay on the ground, or did they come up on the balls of their feet?
- What to do based on the results:
 - **If your arms came forward as you descended & you couldn't maintain or get into an overhead position:**
 - This would send us down a similar path of if you scored a 2 or a 1 on the shoulder mobility assessment. The exercises listed above will help the swimmer build resilient shoulders that can maintain an overhead position.
 - **If their mid back rounded as they descended:**
 - This is a sign they need to work on thoracic extension (when your mid back leans backwards). The best exercises to train this motion will be:
 - Thoracic extension w/ foam roller
 - Thoracic extension w/ overhead raise
 - Thoracic extension w/ pvc press
 - **If their knees came together as they descended:**
 - They can work on the strength of their hip musculature and ankle mobility.
 - To strengthen the hips:
 - Banded hip adduction, copenhagen plank variations, iso hold w/ groin squeeze, side plank w/ hip abduction variations, shin squats, wall supported hip mobility
 - To improve ankle mobility
 - Foam roll the calves, lower leg strengthening prehab series, tissue remodeling series - ankles, calf raises (balls of your feet elevated a few inches off the ground if possible)
 - **If they came up on the balls of their feet as they descended**
 - To improve ankle mobility
 - Foam roll the calves, lower leg strengthening prehab series, tissue remodeling series - ankles, calf raises (balls of your feet elevated a few inches off the ground if possible)

- **3. Seated Hip Internal & External Rotation** - This assessment measures hip internal (when your foot goes out), and external (when your foot goes in) rotation. For breaststrokers/IMers, hip internal rotation especially is essential for a strong & resilient breaststroke kick. But even if you don't swim BR/IM, the mobility of the hips contributes to strength & power in & out of the water. In addition, mobile hips will help you move more efficiently during dryland, which means you will have much more effective dryland workouts. Which will translate to even better results that you can take to the water.
- Generally speaking, from the standpoint of being a well rounded athlete we would like to see at least 35 degrees of both internal & external rotation.

When it comes to gaining & maintaining hip mobility (on top of the hip exercises you do as part of the program), the 2 big factors are 1. addressing the muscles of the area (glutes & groin) and then 2. being consistent with the rotation based hip mobility exercises (like the 90-90 hip switch or heel click). Below are some recommendations that address each of those areas:

- Glutes
 - Foam rolling, stretching, being sure you train with a full range of motion on strength exercises (go as far into the movement as you can without compromising technique)
- Groin
 - Lying hip adduction with band, copenhagen plank level 1, rotational lunges, iso hold w/ groin squeeze, half kneeling adductor w/ rotation
- Hip mobility
 - Wall supported hip mobility, 90-90 hip switch, 90-90 leg sweep, 90-90 heel click, quadruped fire hydrants, side plank w/ hip abduction variations, shin squats
- **4. Half Kneeling Ankle Dorsiflexion** - This assessment will focus on mobility of the ankle joint, without any influence from the hips or upper body. The ankles play the role of creating a strong & powerful kick in the water and also acting as the bridge that transfers force from the athlete's body into the wall on every turn.
 - **If the athlete was unable to touch their knee to the wall without their heel coming up off the ground**
 - To improve ankle mobility
 - Foam roll the calves, lower leg strengthening prehab series, tissue remodeling series - ankles, calf raises (balls of your feet elevated a few inches off the ground if possible)